the opportunity to sample a variety of foods and beverages available for flight. A pack of information is given to each astronaut to use in planning their personal preference menus. Included in the packet is a standard menu, training menu, past flight menus the astronaut has chosen, and the baseline shuttle food and beverage list.

Astronauts select their menu approximately five months before flight. The menus are analysed for nutritional content by the nutritionists and recommendations are made to correct any nutrient deficiencies based on the Recommended Dietary Allowances.

Astronauts eat three meals a day: breakfast, lunch and dinner. Nutritionists ensure the food astronauts eat provides them with a balanced supply of vitamins and minerals. Calorie requirements differ for astronauts. For instance, a small woman would require only about 1,900 calories a day, while a large man would require about 3,200 calories. There are also many types of foods an astronaut can choose from such as fruits, nuts, peanut butter, chicken, beef, seafood, candy, brownies, etc. Drinks range from coffee, tea, orange juice, fruit punches and lemonade.

The menus are then finalized and provided to the Flight Equipment Processing Contractor (FEPC) in Houston three months before launch.

Fitness and exercise

Living in space is not the same as living on Earth. In space, astronauts' bodies change. On Earth, our lower body and legs carry our weight. This helps keep our bones and muscles strong. In space, astronauts float. They do not use their legs much. Their lower backs begin to lose strength. Their leg muscles do too. The bones begin to get weak and thin. This is very bad for astronauts' bodies. So, how do astronauts help their muscles and bones? They must exercise in space every day.

Exercise is an important part of the daily routine for astronauts aboard the station to prevent bone and muscle loss. On average, astronauts exercise two hours per day. The equipment they use is different than what we use on Earth. Lifting 100 kilograms on Earth may be a lot of work, but in space it is easy. Therefore, exercise equipment needs to be specially designed for use in space so astronauts will receive the workout needed. In space, astronauts use three pieces of exercise equipment.

- **Resistance Exercise Device:** The Resistance Exercise Device (RED) simulates weight lifting in microgravity. The system is powered by lightweight, portable SpiraFlex technology, which creates resistance by